

PREPARED FOR: TREVOR KNIGHT, GMD · FOR DISCUSSION WITH MAYOR STEFAN ROOTS

Chester Clean Economy Strategy

AD/RNG + Tire BOO + MaxWellness Health Campus

A revised phased economic development framework for the City of Chester — anchored by a proven, bankable Anaerobic Digestion / Renewable Natural Gas facility as the primary clean tech anchor, a Klean/KBR tire pyrolysis BOO as a secondary revenue stream, and a MaxWellness Community Health Campus at DeShong Park. Revenue-first structure. No City appropriation required.

Merge 4 Design LLC · M4Di™ · Merge4Ward LLC · M4CleanTech LLC · MaxWellness Model

SECTION 1 — REVENUE: WHAT CHESTER EARNS AT STEADY STATE

The Numbers First. The Strategy Follows.

These are ROM gross revenue figures for all Chester platform components at Year 3–5 steady state. All figures are before operating expenses. The purpose of leading with revenue is simple: the conversation with Mayor Roots is not about what Chester spends — it is about what Chester earns, permanently, from land it already controls.

\$4.3M– \$7.9M AD/RNG Facility (Primary) Tipping fees + D3 RIN credits + digestate sales · Year 3 steady state	\$2.9M– \$3.2M MaxWellness Chester — DeShong HCV housing + Medicaid HCBS + clinic + workforce training · Year 3	\$150K– \$300K Tire BOO Ground Lease Spread Klean/KBR sublease net of nominal City rent · Year 1 of BOO ops	\$7.35M– \$11.4M TOTAL GROSS · All Components Annual platform revenue at steady state before operating costs
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Compared to Reworld/Covanta: \$5M/yr host fee — the only economic benefit Chester currently receives — this platform delivers 1.5–2.3× more annual gross revenue, plus 120–210 permanent jobs, brownfield remediation, and 3.9 miles of waterfront unlocked. ROM ±30–40%.

Platform Revenue by Component — ROM Breakdown

REVENUE STREAM	DESCRIPTION	ROM GROSS/YR	TIMING
AD/RNG — Tipping Fees	Organic waste haulers pay to deliver food waste, FOG, and organic MSW. Chester generates large volumes from institutions, restaurants, and food processors. \$60–\$80/ton tipping fee.	\$1.8M–\$3.2M/yr	Year 1 operations

REVENUE STREAM	DESCRIPTION	ROM GROSS/YR	TIMING
AD/RNG — D3 RIN Credits	Federal Renewable Fuel Standard D3 credits for cellulosic biogas upgrading to pipeline RNG. Priced at \$2–\$4/gallon equivalent. Primary federal revenue incentive for AD/RNG projects.	\$1.4M–\$3.1M/yr	Year 1 ops + RIN market
AD/RNG — Digestate Sales	Nutrient-rich soil amendment from AD process. Sold to regional agriculture and composting operations. Secondary but reliable revenue stream.	\$300K–\$600K/yr	Year 2+
AD/RNG Gross Total	Combined tipping fee + D3 RIN + digestate at steady state. D3 RIN pricing creates volatility — stress-test at \$1.50, \$3.00, \$5.00/gal.	\$3.5M–\$6.9M/yr	Year 2–3
RNG Pipeline Sales (Additional)	If pipeline interconnection secured with Enbridge/SJI, RNG injection at \$12–\$18/MMBtu market price adds further upside beyond D3 RIN structure.	+\$400K–\$1.0M/yr	Year 2–3 (optional)
MaxWellness — HCV Housing	60 supportive housing units at average \$1,350/month HCV contract. Section 8 / PBRA revenue. Annually contracted, federally backed.	\$972K/yr	Year 3 (occupancy)
MaxWellness — HCBS / Medicaid	20 IDD/step-down residential beds at \$185/day PA Medicaid HCBS waiver. Stable, recurring Medicaid revenue stream.	\$1.35M/yr	Year 3
MaxWellness — Clinic + Training	Ground floor community medical clinic (FQHC look-alike or CCBHC) plus allied health workforce training with Widener University.	\$600K–\$850K/yr	Year 3
Tire BOO — Ground Lease Spread	M4CleanTech pays City nominal rent. Klean/KBR pays M4CleanTech market industrial ground rent on the waterfront BOO parcel. M4CleanTech pockets the spread. Cash flows Day 1 of operations.	\$150K–\$300K/yr	Year 2 BOO ops
PLATFORM TOTAL — Gross Revenue	All components combined at Year 3–5 steady state. Before operating expenses. MaxWellness NOI est. \$1.1M–\$1.5M/yr after ops.	\$7.35M–\$11.4M/yr	Year 3–5
vs. Reworld Status Quo	Host fee — the only revenue Chester receives from the nation's largest polluter. No jobs. No waterfront access. EJ harm continues.	\$5.0M/yr only	Current

ROM ±30–40%. All figures gross before operating expenses. MaxWellness NOI ~\$1.1M–\$1.5M/yr net at Year 3. AD/RNG NOI ~\$1.2M–\$2.8M/yr at Year 3 after ops. Tire BOO ground lease spread is net (City rent subtracted). D3 RIN revenue is subject to EPA Renewable Fuel Standard policy continuity.

SECTION 2 — TECHNOLOGY DUE DILIGENCE: WHAT WE EVALUATED AND WHY AD/RNG WAS SELECTED

Due Diligence Findings: Technologies Evaluated, AD/RNG Selected

M4CleanTech conducted due diligence across the clean tech processing landscape before selecting AD/RNG as the primary technology anchor for Chester. Plastics pyrolysis was evaluated and rejected following a review of commercial performance across the sector. The findings that governed the technology selection decision are documented below.

PYROLYSIS (EVALUATED — NOT SELECTED)	AD/RNG (REVISED PRIMARY TECHNOLOGY) — WHY BETTER
Plastics pyrolysis: \$260M Brightmark Indiana facility operated at 5% capacity, defaulted on \$172.5M in green bonds, sold for \$14.3M. Declared bankruptcy March 2025.	AD/RNG: Zero analogous commercial failures in US market. Mature technology in use since late 1800s. Thousands of operating facilities globally.
New Hope Energy (Texas), Agilyx (Oregon), Ionika (Netherlands) all closed or ceased operations 2024–2025. Lux Research: industry tracking less than half of projected capacity.	Linden NJ AD/RNG facility (\$440M) operational on 21-acre brownfield — exact Chester profile. 3,783 MMBtu/day. Bloom Energy fuel cell on-site. Template project.
Tire pyrolysis (TDP) more defensible than plastics — homogeneous feedstock, real recovered carbon black market. But still no proven large-scale US commercial TDP deployment.	D3 RIN credits (federal Renewable Fuel Standard) create a real, liquid, tradeable revenue stream with no equivalent in pyrolysis. Banks understand and underwrite this.
EJ community opposition infrastructure (CRCQL, Beyond Plastics, Energy Justice Network) is organized and effective. Any thermal facility in Chester faces 12–18 month community consent process minimum.	Food waste diversion is universally supported EJ narrative — reduces landfill methane harming EJ communities. Chester goes from fighting industrial harm to leading food waste recovery.
Pyrolysis: harder lender conversation post-Brightmark. CDFI and mission banks pulling back from pyrolysis-only projects. Green bond defaults created lender skepticism.	AD/RNG: CDFI and mission banks actively financing. Reinvestment Fund, Artisans' Bank, CDFI Coal (PA) all have active AD/RNG portfolios. Chester is a strong application.

TIRE BOO RETAINED AS SECONDARY REVENUE

Tire pyrolysis via Klean/KBR BOO remains in the Chester strategy as a secondary revenue component on a separate waterfront parcel. The BOO structure means Klean/KBR funds, owns, and operates the facility — M4CleanTech provides site access and pockets ground lease spread with zero capital exposure. If Klean/KBR executes, great. If they don't, AD/RNG carries the entire financial thesis independently. The tire BOO is upside, not load-bearing.

SECTION 3 — AD/RNG: TECHNOLOGY, FEEDSTOCK, AND CHESTER FIT

Anaerobic Digestion: Proven at Scale, Bankable, Community-Positive

Anaerobic digestion is the biological decomposition of organic material in sealed, oxygen-free tanks. The process produces biogas (50–70% methane) which is upgraded to pipeline-quality Renewable Natural Gas, plus digestate (nutrient-rich soil amendment sold to agriculture). The technology has operated commercially since the late 1800s. There is no scaling uncertainty, no pilot risk, and no independent verification gap. The Linden, NJ facility — a \$440M AD/RNG project on a 21-acre brownfield in northern New Jersey — is the direct template for Chester.

COMPONENT	DESCRIPTION	ROM QUANTITY/VALUE	KEY MILESTONE
Chester Food Waste Feedstock	Philadelphia metro + Delaware County institutions, restaurants, food processors, and grocery chains. More than 75% of regional food waste currently landfilled — available for diversion at competitive tipping fees.	150,000–300,000 tons/yr available	Signed tipping contracts before lender commitment
FOG (Fats, Oils, Grease)	Restaurant grease traps and food processing FOG streams. High-energy feedstock that boosts biogas yield significantly. Regional haulers actively seeking licensed AD tipping sites.	Boosts RNG yield 25–40%	Direct hauler contracts
Biogas Production	150,000 tons/yr organic feedstock at typical 65 m ³ /ton yield = approx. 9.75 million m ³ biogas/yr, upgraded to approximately 4.5–5.5 million m ³ RNG.	~4.5M–5.5M m ³ RNG/yr	Year 2 steady state
D3 RIN Revenue	Food waste AD qualifies for D3 (cellulosic) RIN classification — highest-value RIN category under federal RFS. Each mmBtu of RNG generates approximately 11.7 D3 RINs.	\$1.4M–\$3.1M/yr	RIN price \$2–\$4/gal equiv.
Pipeline Interconnection	Chester has Enbridge / South Jersey Industries gas infrastructure. Pipeline injection agreement + interconnection engineering is the critical path item — typically 18–24 months from project initiation.	\$500K–\$1.5M CapEx	Year 1–2 development
Facility Footprint	Typical 100–200 ton/day AD/RNG facility: 3–6 acres. Fits Reworld waterfront parcel or adjacent DCIDA industrial site. Phase 1 at 100 ton/day with expansion to 200 ton/day designed in.	3–6 acres Phase 1	Expandable to 6–10 acres

SECTION 4 — MAXWELLNESS CHESTER AT DESHONG PARK

DeShong: 5.2 Developable Acres, C-1/R Zoning, MaxWellness Model Ready

The DeShong Park site overlay requires 60% conservation easement (7.8 acres preserved) — leaving 5.2 acres developing under C-1/R commercial standards. The MaxWellness compact mid-rise model, already in development for Claymont DE (AE25-020), adapts directly to this footprint. Program, financing, and operating model are proven at the component level.

ZONE	USE & REVENUE SOURCE	SF (ROM)	ANNUAL REVENUE ROM
Zone A — Ground Floor	Community Medical / Behavioral Health Clinic. FQHC look-alike or CCBHC designation unlocks federal operating grants.	8,000–10,000 SF	\$420K–\$600K/yr
Zone A — Upper Floors	60 supportive housing units. HCV/Section 8 or PBRA contract. PA Housing Choice Voucher program.	48,000–56,000 SF	\$972K/yr
Zone B	20 IDD/step-down residential beds. PA Medicaid HCBS / ODP waiver revenue. Stable Medicaid income stream.	24,000–32,000 SF	\$1.35M/yr
Zone C	Allied health workforce training. Widener University partnership. WIOA + DCED grant-eligible.	6,000–8,000 SF	\$180K–\$250K/yr
TOTAL — MaxWellness Chester	80–100 residential units. Built with Forward Building System — CLT/SIP panels from Merge4Ward Wilmington node.	~86,000–106,000 SF	\$2.9M–\$3.2M gross/yr
Total Project Cost ROM	PHFA LIHTC + HOME + CDFI + DCED RACP + philanthropic gap. Construction: 180–240 jobs (40% Merge4Ward Forward Building System).	\$20.5M–\$25.8M	NOI est. \$1.1M–\$1.5M/yr net

SECTION 5 — PLATFORM FINANCE STACK

Grant-Heavy. No City Appropriation. Outside Capital Does the Work.

TRANCHE	SOURCE & RATIONALE	ROM AMOUNT	COST	STRUCTURE
EPA Brownfields Assessment + Cleanup	Waterfront site brownfield qualification. Funds Phase I/II ESA + remediation. EJ community overlay strengthens application.	\$500K–\$1.5M	0% — grant	Non-dilutive
EPA EJ Thriving Communities	Chester is textbook EJ community. Food waste diversion replacing incinerator is strongest possible EJ Thriving Communities use case.	\$1M–\$3M	0% — grant	Non-dilutive
EDA Economic Adjustment Assistance	Chester qualifies as distressed community. AD/RNG + MaxWellness job creation + site remediation all EAA-eligible.	\$1M–\$3M	0% — grant	Non-dilutive
PHFA LIHTC (MaxWellness)	PA Housing Finance Agency Low Income Housing Tax Credit equity for MaxWellness Chester housing component.	\$6M–\$9M equity	0% — tax credit	Non-dilutive equity
PA DCED RACP	Redevelopment Assistance Capital Program. AD/RNG infrastructure + MaxWellness both RACP-eligible.	\$1M–\$5M	0% — grant	Non-dilutive
CCBHC Federal Grant (Clinic)	Certified Community Behavioral Health Clinic designation unlocks federal operating grants for ground floor clinic.	\$1M–\$2M/yr	0% — federal operating	Annual recurring
CDFI Senior Debt — AD/RNG	Reinvestment Fund, NCALL, or PA-based CDFI. AD/RNG project finance is well-understood by CDFI lenders. Tipping contracts + D3 RINs support underwriting.	\$4M–\$8M	4–6%	15–20 yr term
Klean/KBR BOO (Tire Facility)	Klean/KBR funds, owns, and operates tire pyrolysis facility under BOO sublease. Zero M4D capital required for clean tech component.	\$8M–\$15M operator-funded	BOO operator	No M4D capital
TOTAL PLATFORM — All Sources	AD/RNG + MaxWellness + Tire BOO combined. M4D capital exposure below \$500K assuming grant stack executes.	\$22.5M–\$46.5M	Blended 2–3%	M4D exposure <\$500K

ROM ±30–40%. Grant amounts depend on application quality and competitive cycle timing. PHFA LIHTC subject to annual allocation round. CDFI loan subject to lender underwriting at time of application. BOO economics subject to Klean/KBR commercial negotiation.

SECTION 6 — IMPLEMENTATION TIMELINE

Three Horizons. Each Builds on the Last.

NOW — Q4 2026 Positioning & Agreements Ground lease + LOIs Exclusive Development Agreement with City/DCIDA on Reworld waterfront. M4CleanTech contacts Bioenergy Devco, RNG Energy Solutions, Vanguard Renewables for AD/RNG BOO or development partnership. DCIDA engagement on DeShong RFP status. M4Di G0 ROM locked on both sites. Feedstock LOIs from regional haulers initiated.	Q1–Q3 2027 Phase 1A Feasibility 12–14 weeks · grant-fundable M4Di Phase 1A on waterfront AD/RNG site and DeShong. Feedstock contracts executed. Pipeline interconnection pre-application with Enbridge/SJI. PHFA LIHTC pre-application. EPA EJ + EDA EAA grant applications with Mayor endorsement. PA DEP pre-application meeting for AD/RNG facility.	2027–2029 Design & Finance Close Capital stack assembled AD/RNG facility design + PA DEP permit. MaxWellness Chester design (M4D AOR). PHFA LIHTC award. CDFI loan commitment. AD/RNG construction commences. Klean/KBR BOO sublease executed on second waterfront parcel. Merge4Ward Wilmington panels supply MaxWellness construction.	2029–2032+ Operations & Steady State \$7.35M–\$11.4M gross/yr AD/RNG operational. MaxWellness Chester opens — 80–100 units, clinic, training. Tire BOO Phase 1 operational. Chester waterfront corridor activated. D3 RIN revenue flowing. CRCQL partnership sustained. Mayor Roots' EJ reinvestment legacy established.
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SECTION 7 — WHAT WE ARE ASKING MAYOR ROOTS

Three Authorizations. No Budget Required.

Authorization 1 — Exclusive Development Agreement on Reworld Waterfront

Direct DCIDA to negotiate an EDA with M4CleanTech on the Reworld waterfront parcel — exclusive right to develop an AD/RNG facility and tire BOO for 18–24 months pending Reworld's departure. Cost to City: zero. Effect: positions M4CleanTech ahead of any competing developer before the site is publicly marketed. Must include Right of First Refusal on adjacent parcels.

Authorization 2 — Introduction to Finance / Economic Development Team

Facilitated meeting with Chester's finance director and DCIDA contacts to walk through ROM revenue figures in this brief, stress-test the capital stack, and begin identifying which grant programs the City can actively pursue alongside M4CleanTech. This is a working session — we want to be corrected where we are wrong.

Authorization 3 — Letters of Support for AD/RNG and Klean/KBR Outreach

Letters of support from Mayor Roots for M4CleanTech's outreach to Bioenergy Devco / RNG Energy Solutions (AD/RNG development partners) and Klean Industries/KBR (tire BOO). The Mayor's endorsement transforms cold technology partnership inquiries into City-sponsored conversations. Trevor is authorized to present these letters in introductory meetings.

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All ROM investment, revenue, and savings figures are preliminary estimates with ±30–40% confidence range. Not guaranteed, not funded commitments, and not GMP. Produced for concept-level conversation and to frame the scope of M4Di predevelopment engagement. Revenue figures are gross before operating expenses. AD/RNG revenue depends on D3 RIN pricing, tipping fee contracts, and pipeline interconnection agreement terms — all require negotiated execution before financial close. Ground lease, BOO sublease, and supplier exclusivity arrangements require legal review. Grant capital is subject to competitive application and lender underwriting. Pennsylvania regulatory compliance requires PA-licensed counsel. DNREC environmental review and DEDO approvals are conditions precedent to any DE construction commitment.